DIFFERENTIABILITY

SYNOPSIS

Differentiability at a point :

- (i) A function f(x) is differentiable at a point x = a, if $\underset{x\to a}{Lt}\frac{f(x)-f(a)}{x-a}$ exists finitely and it is denoted by $f'(a)$

i.e $f'(a)=\underset{x\to a}{Lt}\frac{f(x)-f(a)}{x-a}$

(ii) The right hand derivative of f(x) at x = a is denoted by $f'(a+)$ and is defined as

i.e $f'(a+)=\underset{h\to 0}{Lt}\frac{f(a+h)-f(a)}{h}$

(iii) The left hand derivative of $f(x)$ at x = a is denoted by $f'(a-)$ and is defined as

i.e $f'(a-)=\underset{h\to 0}{Lt}\frac{f(a-h)-f(a)}{-h}$

W.E.-1 : *The differentiablity of*

$$f(x)=\begin{cases} x\left(\frac{e^{1/x}-e^{-1/x}}{e^{1/x}+e^{-1/x}}\right), & x\neq 0 \\ 0, & x=0\end{cases}$$

at x = 0

Sol: We have $f'(0^-)=\lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$

$$=\lim_{h\to 0}\left(\frac{e^{-2/h}-1}{e^{-2/h}+1}\right)=\frac{0-1}{0+1}=-1$$

Similarly $f'(0^+)=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}=1$

$\therefore$ LHD $\neq$ RHD

$\therefore$ f is not differentiable at x = 0

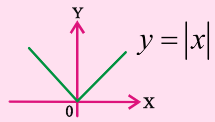
(iv) If f is differentiable at x = a then f is also continous at $x=a$. However the converse need not be true.

(v) If f is not continous at x = a then f is not differentiable at x = a

W.E-2: *$f(x)=\sin x$ is differentiable at x= 0 $\Rightarrow$ f is continuous at x = 0*

W.E.-3 : *Examine the continuity and differentiability of $f(x)=|x|$ at x = 0*

Sol:



Hence $y=|x|$ is continuous everywhere but not differentiable at x = 0

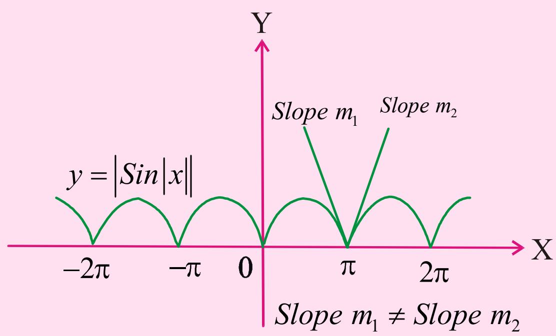
($\because$ sharp corner at x = 0)

W.E-4: *Examine the continuity and differentiability of $y=|\sin|x||$ at $x=n\pi, n\in I$*

Sol: It is clear from the graph that $y=|\sin|x||$ is continuous everywhere but not differentiable at $x=.....-2\pi,\ -\pi,\ 0,\ \pi, 2\pi,..........$

i.e., $x=n\pi, n\in I$

We observe that at all integral values of π, f has a sharp corner



Differentiability of a function over an interval :

- i) A function $f(x)$ defined on an (a,b) is said to be differentiable in (a,b) if it is differentiable at each point of (a,b) ii)A function $f(x)$ defined on [a,b] is said to be differentiable or derivable if

a) f is differentiable from the right at a.

b) f is differentiable at every point on (a,b)

c) f is differentiable from the left at b.

iii)A function f is said to be a differentiable function, if it is differentiable at every point on its domain.

iv)Exponential, logarithemic, trigonometric, inverse trigonometric functions are differentiable in their domain.

v)Polynomial, constant functions are differentiable at each point 'x', where $x \in R$

Standard Results

$f(x)$	$g(x)$	$f(x) \pm g(x)$	$f(x).g(x)$
Differentiable	Differentiable	Differentiable	Differentiable
Differentiable	Non Differentiable	Non Differentiable	May bo or not
Non Differentiable	Non Differentiable	May bo or not	May bo or not

i) $|x-a|$ is not diffrentiable at x = a

ii) $(x-a)^n|x-a|$ is diffrentiable when $n \geq 1$ and is not differentiable when n < 1

iii) $Sgn(x-a)$ is not differentiable at x = a

iv) $x^n \sin\frac{1}{x}, x^n \cos\frac{1}{x}$ are differentiable when n>1 and are not differentiable when $n \leq 1$

v) $\{x\}, [x]$ are not differentiable at all intergral points of x

vi) $\frac{d}{dx}\int_{\alpha(x)}^{\beta(x)} f(t)dt = f(\beta(x))\frac{d}{dx}\beta(x) - f(\alpha(x))\frac{d}{dx}\alpha(x)$

W.E.-5 : Let f:R→R is differentiable function & f(1)= 4 then $g(x) = \underset{x\to 1}{Lt}\int_4^{f(x)} \frac{2t}{x-1}dt =$

Sol : $g(x) = \lim_{x\to 1}\int_4^{f(x)} \frac{2t}{x-1}dt = \lim_{x\to 1}\frac{\int_4^{f(x)} 2tdt}{x-1}$

Apply L-Hospital rule

$= 2\lim_{x\to 1}\frac{f(x)f'(x) - 4.0}{1}$

$= \lim_{x\to 1} 2f(x)f'(x) = 2f(1)f'(1) = 8f'(1)$

Differentiability of Functional Equations

➢ (i) if f(x + y) = f(x).f(y) then

$f'(x) = f'(0).f(x)$

Proof: $f'(x) = \underset{h\to 0}{Lim}\frac{f(x+h) - f(x)}{h}$

$= \underset{h\to 0}{Lim}\frac{f(x)(f(h)-1)}{h}$

$= f(x).f'(0)$

(ii) Funtional equation relations.

a) $f(x+y) = f(x).f(y) \forall x, y$

$\Rightarrow f(x) = a^x (a > 0)$

b) $f(x+y) = f(x) + f(y) \ \forall x, y \in R$

$\Rightarrow f(x) = kx$

c) $f(xy) = f(x).f(y) \forall x, y \in R$

$\Rightarrow f(x) = x^n$

d) $f(xy) = f(x) + f(y) \forall x, y \in R^+$

$\Rightarrow f(x) = k \log x (x > 0)$

e) $f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) \forall x \in R - \{0\}$

$\Rightarrow f(x) = 1 \pm x^n$

(f) $f\left(\frac{mx+ny}{m+n}\right) = \frac{m f(x) + n f(y)}{m+n}$, $m+n \neq 0$

$\Rightarrow f(x) = ax + b$

W.E-6: ***If $f(x+y) = f(x).f(y)\ \forall x, y \in R$, $f(5) = 2$, $f'(0) = 3$ then $f'(5) =$ [AIEEE 2002]***

Sol: $f'(x) = f'(0).f(x)$

$f'(5) = f'(0).f(5) = (3)(2) = 6$

➢ **Vertical tangent :** It may happen that a function is continuous and smooth (with no sharp point) at x = a, even though function is non-differentiable at x=a. This happens due to vertical tangent at $P(a, f(a))$. For vertical tangent line at $P(a, f(a))$, $f(x)$ should be continuous at $x = a$ and $|f(x)| \to \infty$ as $x \to a$

➢ **Oscillation Point :** If a function $f(x)$ is continuous but left and right derivative do not exist at x = a due to high frequency oscillations in neighbourhood of x = a, then the function $f(x)$ is non-differentiable at x = a and such a point is called oscilation point.

C.U.Q

1. ***If both $f(x)$ & $g(x)$ are differentiable functions at $x = x_0$, then the function defined as, $h(x) =$ Maximum $\{f(x), g(x)\}$:***
   1) is always differentiable at $x = x_0$
   2) is never differentiable at $x = x_0$
   3) is differentiable at $x = x_0$ when $f(x_0) \neq g(x_0)$
   4) can not be differentiable at $x = x_0$ if $f(x_0) = g(x_0)$.

2. ***If f(x) differentiable everywhere then***
   1) $|f(x)|$ is differentiable everywhere
   2) $|f|^2$ is differentiable everywhere
   3) $[f(x)]$ is differentiable everywhere
   4) $Sgn(f(x))$ is differentiable everywhere

3. ***If $f(x) = (x - a)\,g(x)$ and $g(x)$ is continuous at $x = a$, then $f'(a) =$***
   1) a g(a) 2) g'(a) 3) g(a) 4) ag'(a)

4. ***If $f(x) = sgn(x)$ then $f'(0) =$***
   1) 0 2) -1
   3) 1 4) does not exist

5. ***If $f(0) = 0$, $f'(0) = 2$ then derivative of $y = f(f(x))$ at $x = 0$ is***
   1) 1 2) 2 3) 4 4) 8

C.U.Q - KEY

1) 3 2) 2 3) 3 4) 4 5) 3

C.U.Q - HINTS

1. Consider the graph of $h(x) = \max(x, x^2)$ at $x = 0$ and $x = 1$
   for Differentiability : $h(x) = \max.(x^2, -x^2)$
2. Conceptual.
3. By using $f(a) = 0$, we get $f'(a) = g(a)$
4. $f(x) = \begin{cases} 0, & x = 0 \\ -1, & x < 0 \\ 1, & x > 0 \end{cases}$ f(x) is not continuous at $x=0$
5. $\frac{dy}{dx} = f'(f(x)).f'(x)$

LEVEL - I (C.W)

Differentiability at a point:

1. ***Let $f(x) = x^{3/2} - \sqrt{x^3 + x^2}$ then***
   1) LHD at $x=0$ exist but RHD at $x=0$ does not exist
   2) $f(x)$ is not differentiable at $x=0$
   3) RHD at $x=0$ exist but LHD at $x=0$ does not exist
   4) $f(x)$ is differentiable and continuous at $x=0$

2. ***If $f(x) = x\left(\frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}\right)$, $x \neq 0\ (a>0)$, $f(0)=0$***

then

1) f is differentiable at $x=0$

2) f is not differentiable at $x=0$

3) f is not continuous at $x=0$

4) $\lim_{x\to 0} f(x)$ does not exists

3. Let f(x)= $\begin{cases} x^2, & if\ x \le x_0 \\ ax+b, & if\ x > x_0 \end{cases}$. If f is differentiable at x_0 then

1) $a = x_0, b = -x_0$ 2) $a = 2x_0, b = -x_0^2$

3) $a = 2x_0, b = x_0^2$ 4) $a = x_0, b = x_0^2$

4. *Let f(x)=|x-a| ϕ (x), where ϕ is a continuous function and $\phi(a) \neq 0$. Then*

1) $f'(a+) = \phi'(a)$ 2) f is differentiable at $x=a$

3) $f'(a+) = -\phi'(a)$ 4) $f'(a-) = -\phi(a)$

5. *The left-hand derivative of f(x)=[x]sin πx at x = k, k is an integer is*

1) $(-1)^k(k-1)\pi$ 2) $(-1)^{k-1}(k-1)\pi$

3) $(-1)^k k\pi$ 4) $(-1)^{k-1}k\pi$

6. *If* $f(x) = \begin{cases} 1 & x < 0 \\ x^2 & x \ge 0 \end{cases}$ *then at* $x=0$

1) f' is differentialble 2) f is continuous

3) f is differentiable 4) f is not continuous

7. *If f(x)=|x - a |+| x + b|, $x \in R$, b>a>0. Then*

1) $f'(a+) = 1$ 2) $f'(a+) = 0$

3) $f'(-b+) = 0$ 4) $f'(-b+) = 1$

Differentiability over an interval:

8. *Let* $f(x) = \begin{cases} x & x < 1 \\ 2-x & 1 \le x \le 2 \\ -2+3x-x^2 & x > 2 \end{cases}$ *then f(x) is*

1) differentiable at $x=1$

2) differentiable at $x=2$

3) differentiable at $x=1$ and $x=2$

4) not differentiable at $x=0$

9. *Let* $f(x) = \begin{cases} x^2/2 & 0 < x \le 1 \\ 2x^2 - 3x + 3 & 1 < x < 2 \end{cases}$ *then which is correct*

1) f is continuous in [0,2]

2) f' is continuous in [0,2]

3) $f(x)$ is discontinuous at $x = 1$

4) $\underset{x\to 1+}{Lt} f(x) = \underset{x\to 1-}{Lt} f(x)$

10. *Let f(x)=asin|x|+be$^{|x|}$ is differentiable when*

1) a = -b 2) a = b 3) a = 0 4) b = 0

LEVEL - I (C.W)-KEY

1) 3	2) 2	3)2	4) 4	5) 1
6) 4	7) 3	8) 2	9) 3	10) 1

LEVEL - I (C.W)-HINTS

1. $f(x) = x^{3/2} - \sqrt{x^3 + x^2}$

$\therefore$ L.H.D at $x=0$ does not exist as $D_f = [0, \infty)$

2. $\lim_{x\to 0+} f(x) = \lim_{x\to 0+} \frac{x(a^{1/x} - a^{-1/x})}{a^{1/x} + a^{-1/x}} = \lim_{x\to 0+} \frac{x(1 - a^{-2/x})}{1 + a^{-2/x}} = 0$

also, $\lim_{x\to 0-} f(x) = \lim_{x\to 0-} x\left(\frac{a^{2/x} - 1}{a^{2/x} - 1}\right) = 0$

so f is continuous at $x=0$

$f'(0+) = \lim_{x\to 0+} \frac{h(a^{1/h} - a^{-1/h})}{h(a^{1/h} + a^{-1/h})} = \lim_{x\to 0+} \frac{1 - a^{-2/h}}{1 + a^{-2/h}} = 1$

similarly $f'(0-) = -1$ hence ,f is not differentiable at $x=0$

3. Since f is differentiable so it is continuous also,

$= \lim_{h\to 0} \frac{a+b-h+h-a-b}{h} = 0$

therefore $x_0^2 = f(x_0) = \lim_{x\to x_0^+} f(x) = ax_0 + b$

also, $\lim_{h\to 0+} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h\to 0+} \frac{a(x_0+h) + b - x_0^2}{h}$

$= \lim_{h\to 0+} \frac{x_0^2 + ah - x_0^2}{h} = a\ (\because x_0^2 = ax_0 + b)$

thus $a = f'(x_0 -) = \lim_{h\to 0+} \frac{(x_0+h)^2 - x_0^2}{h} = 2x_0$

hence $x_0^2 = 2x_0^2 + b$ then $b = -x_0^2$

4. $f'(a+) = \lim_{h\to 0+} \frac{|a+h-a|\phi(a+h) - 0}{h}$

$= = \lim_{h\to 0+} \frac{|h|\phi(a+h)}{h} = \lim_{h\to 0+} \phi(a+h) = \phi(a)$

similarly $f'(a-) = -\phi(a)$

5 . Clearly f(k)=0, so the left hand derivative is equal to $\lim_{h\to 0-}\frac{f(k+h)-f(k)}{h}$

$=\lim_{h\to 0-}\frac{[k+h]\sin(k+h)\pi}{h}=\lim_{h\to 0-}\frac{(k-1)\sin(k\pi+h\pi)}{h}$

$=\lim_{h\to 0-}\frac{(k-1)(-1)^k\sinh\pi}{h}$ (since $h<0$) $=(k-1)(-1)^k\pi$

6. $f(0^-)\neq f(0^+)$

7. $f'(x)=\begin{cases}-2, & if\ x<-b\\ 0, & if\ -b\le x\le a\\ 2, & if\ x>a\end{cases}$

$f'(a+)=2,\ f'(-b+)=0$

8. $f'(x)=\begin{cases}1, & if\ x<1\\ -1, & if\ 1\le x\le 2\\ 3-2x, & if\ x>2\end{cases}$

$f'(1-)=1,\ f'(1+)=-1,\ f'(2-)=-1,\ f'(2+)=-1$

$f'(2-)=f'(2-)$

9. $\underset{x\to 1-}{Lim} f(x)=1/2$

$\underset{x\to 1+}{Lim} f(x)=2$

10. f(x)=$\begin{cases}-a\sin x+be^{-x} & if\ x<0\\ a\sin x+be^{x} & if\ x>0\end{cases}$

$f'(x)=\begin{cases}-a\cos x-be^{-x} & if\ x<0\\ a\cos x+be^{x} & if\ x>0\end{cases}$

then $f'(0-)=-a-b$ and $f'(0+)=a+b$

If a=-b, then $f'(0-)=f'(0+)$

LEVEL - I (H.W)

Differentiability at a point:

1. Let f(x)=|x-1|+|x+1|

1) f(x) is differentiable at $x=\pm 1$

2) f(x) is not differentiable at $x=\pm 1$

3) f(x) is neither continuous nor differentiable at $x=\pm 1$

4) f(x) is not continuous at x=0

2. Let $f(x)=\begin{cases}0 & \forall x<0\\ x^2 & \forall x\ge 0\end{cases}$ **then** $\forall x\in R$

1) f is differentiable

2) f,f′ both are continuous

3) f is differentiable, f′ is continuous

4) All of these

3. Let $f(x)=\begin{cases}\frac{x}{1+2^{1/x}}, & x\neq 0\\ 0 & x=0\end{cases}$ **then**

1) LHD f(x) at x=0 is 1

2) RHD of f(x) at x=0 is not equal to zero

3) f(x) is differentiable at x=0

4) $\underset{x\to 0}{Lt} f(x)=1$

4. If f(x)=|$\log_{10}$x| then at x=1

1) f is not continuous

2) f is continuous but not differentiable

3) f is differentiable

4) the derivative is 1

5. $f(x)=\begin{cases}|x-4| & for\ x\ge 1\\ \frac{x^3}{2}-x^2+3x+\frac{1}{2} & for\ x<1\end{cases}$**, then**

1) f(x) is continuous at x=1 and x=4

2) f(x) is differentiable at x=4

3) f(x) is continuous and differentiable at x=1

4) f(x) is only continuous at x=1

6. If $f'(3)=2$, then $\underset{h\to 0}{Lim}\frac{f(3+h^2)-f(3-h^2)}{2h^2}$ is

1) 1 2) 2 3) 3 4) 1/2

7. If [.] denote the greatest integer function and $f(x)=[\tan^2 x]$, then

1) $\underset{x\to 0}{Lim} f(x)$ does not exist

2) f is not continuous at $x=0$

3) $f(x)$ is differentiable at $x=0$

4) $f'(0)=1$

8. If $f(x)=|x|e^x$, then at $x=0$

1) f is continuous

2) f is continuous but not differentiable

3) f is differentiable

4) the derivative is 1

Differentiability over an interval:

9. The set of all points where the function

$f(x)=\sqrt{1-e^{-x^2}}$ **is differentiable is**

1) $(0,\infty)$ 2) $(-\infty,\infty)$

3) $(-\infty,\infty)-\{0\}$ 4) $(-\infty,\infty)-\{0,1,2\}$

10. **The set of all points where** $f(x)=2x|x|$ **is differentiable**

1) $(-\infty,\infty)$ 2) $(-\infty,\infty)-\{0\}$

3) $(0,\infty)$ 4) $[0,\infty)$

LEVEL-I (H.W)-KEY

1) 2	**2) 4**	**3)1**	**4) 2**	**5) 1**
6) 2	**7) 3**	**8) 2**	**9) 3**	**10) 1**

LEVEL-I (H.W)-HINTS

1. $f(x)=\begin{cases}-2x, \text{ if } x<-1\\ 2, \text{ if } -1\le x<1\\ 2x, \text{ if } x\ge 1\end{cases} \Rightarrow f'(x)=\begin{cases}-2, \text{ if } x<-1\\ 0, \text{ if } -1\le x<1\\ 2, \text{ if } x\ge 1\end{cases}$

$f'(-1^-)=-2,\ f'(-1^+)=0,\ f'(1^-)=0,\ f'(1^+)=2$

$\therefore$ f(x) is not differentiable at $x=\pm 1$

2. $f(x)=\begin{cases}0 & \forall x<0\\ x^2 & \forall x\ge 0\end{cases} \Rightarrow f'(x)=\begin{cases}0 & \forall x<0\\ 2x & \forall x\ge 0\end{cases}$

then $f''(x)=\begin{cases}0 & \forall x<0\\ 2 & \forall x\ge 0\end{cases}$

$f'(0-)=f'(0+)=0$ *and*

$f''(0-)=0, f''(0+)=2$

$\therefore$ f and f' are differentiable

3. $f'(0-)=\lim\limits_{x\to 0^-}\dfrac{f(x)-f(0)}{x-0}=\lim\limits_{x\to 0^-}\dfrac{\frac{x}{1+2^{1/x}}-0}{x}$

$=\dfrac{1}{1+2^{1/0-}}=\dfrac{1}{1+2^{-\infty}}=1$

$f'(0+)=\lim\limits_{x\to 0+}\dfrac{f(x)-f(0)}{x-0}=\lim\limits_{x\to 0+}\dfrac{\frac{x}{1+2^{1/x}}-0}{x}$

$=\dfrac{1}{1+2^{1/0+}}=\dfrac{1}{1+2^{\infty}}=0$

$\therefore$ L.H.D of f(x) at x=0 is 1

4. Since g(x)=|x| is continuous but not differentiable at x=0, so f is continuous but not differentiable for $\log_{10}x=0$ i.e for x=1

5. Since g(x)=|x| is a continuous function and $\lim\limits_{x\to 1+} f(x)=3=\lim\limits_{x\to 1-} f(x)$, so f is continous function. In particular f is continuous at x=1 and x=4) f is clearly not differentiable at x=4) Since g(x)=|x| is not differentiable at x=0. Now

$f'(1+)=\lim\limits_{h\to 0+}\dfrac{f(1+h)-f(1)}{h}=\lim\limits_{h\to 0+}\dfrac{|-3+h|-3}{h}=-1$

$f'(1-)=\lim\limits_{h\to 0-}\dfrac{(1/2)(1+h)^3-(1+h)^2+3(1+h)+(1/2)-3}{h}$

$==\lim\limits_{h\to 0-}\dfrac{(1/2)(h^3+3h^2+3h)-(h^2+2h)+3h}{h}=\dfrac{5}{2}$

6. $\lim\limits_{h\to 0}\left(\dfrac{f(3+h^2)-f(3-h^2)}{2h^2}\right)$

$=\dfrac{1}{2}\left\{\lim\limits_{h\to 0}\dfrac{f(3+h^2)-f(3)}{h^2}-\lim\limits_{h\to 0}\dfrac{f(3-h^2)-f(3)}{h^2}\right\}$

$=\dfrac{1}{2}\{f'(3)-(-f'(3))\}$

$=\dfrac{1}{2}\times 2f'(3)=2$

7. $0<x<\pi/4$, $[\tan^2 x]=0$. Also $\tan^2 x$ is an even function

$\lim\limits_{x\to 0+}\left(\dfrac{f(0+x)-f(0)}{x}\right)$

$=\lim\limits_{h\to 0}\dfrac{f(h)-f(0)}{h}=0$

$\therefore$ f is continuous at x = 0 and differentiable x = 0. Also $f'(0)=0$

8. $L.H.D=\lim\limits_{x\to 0-}\dfrac{-xe^x}{x}=-1$

$R.H.D=\lim\limits_{x\to 0^+}\dfrac{xe^x}{x}=1$

9. $f'(x)=\dfrac{xe^{-x^2}}{\sqrt{1-e^{-x^2}}}$ is not differentiable only at x = 0

10. $f(x)=\begin{cases}2x^2 & x\ge 0\\ -2x^2 & x<0\end{cases}$ is differentiable everywhere.

LEVEL -II (C.W)

Differentiability at a point:

1. If $f(x)=\begin{cases}-\frac{1}{2}x^2, & for \quad x<1\\ \frac{3}{2}x^2+1, & for\ x\ge 1\end{cases}$, then

1) f is differentiable everywhere on R
2) $f'(1-)=-1$ and $f'(1+)=0$
3) $f'(1-)=-1$ and $f'(1+)=3$
4) $f'(1-)=1$ and $f'(1+)=-1$

2. If $f(x)=\frac{x\sqrt{x}}{x+2}$ then derivative of $f(x)$ at $x=0$ is

1) 0 2) 1 3) 1/2 4) does not exist

3. Let f and g be differentiable functions satisfying $g(a)=b, g'(a)=2$ and fog=I(identity function). Then $f'(b)$ is equal to

1) $1/2$ 2) 2 3) $2/3$ 4) 3

4. Let $f(x)=x^2|x|$ then the set of values where f(x) is three times differentiable is

1) Infinite 2) 2 3) 3 4) 0

5. If $f(x)=p|\sin x| + qe^{|x|}+r|x|^3$ and f(x) is differentiable at x=0, then

1) q+r=0; p is any real number
2) p+q=0; r is any real number
3) q=0, r=0; p is any real number
4) r=0,p=0; q is any real number

6. If $f(x)=\begin{cases}-3x+2, & x<1\\ \frac{1}{2}x^2+7, & x\ge 1\end{cases}$, then which of the following is not true

1) $f'(1+)=1$ 2) $f'(1-)=-3$
3) $f'(1-)=f'(1+)=1$
4) f is not differentiable at x = 1

7. Let $f(x)=\frac{\sin 4\pi[x]}{1+[x]^2}$, where $[x]$ is the greatest integer less than or equal to x, then

1) f(x) is not differentiable at some points
2) f(x) exists but is different from zero
3) LHD (at x = 0) = 0, RHD (at x = 1) = 0
4) $f'(x)=0$ but f is not a constant function

8. The function given by $y=||x|-1|$ is differentiable for all real numbers except the points

1) $\{0,1,-1\}$ 2) ± 1 3) 1 4) -1

9. If $f:R\to R$ is an even function which is twice differentiable on R and $f''(\pi)=1$ then $f''(-\pi)$ is

1) -1 2) 0 3) 1 4) 2

Differentiability over an interval:

10. Let a function y = f(x) be difined as $x=2t-|t|, y=t^2+t|t|$, Where $t\in R$. then f(x) is

1) Continuous and differentiable in $[-1,1]$
2) Continuous but not differentiable in $[-1,1]$
3) Continuous $[-1,1]$ & differentiable in (-1,1) only
4) Discontinuous on $[-1,1]$

11. Let $f(x)=\begin{cases}-1, & -2\le x<0\\ x^2-1, & 0<x\le 2\end{cases}$ and $g(x)=|f(x)|+f|x|$ then the number of points which g(x) is non differentiable, is

1) at most one point 2) 2
3) exactly one point 4) infinite

12. If $f(x)=\begin{cases}[\cos \pi x], & x<1\\ |x-2|, & 1\le x<2\end{cases}$, then f(x) is

1) discontinuous and non-differentiable at x=-1 and x=1
2) continuous and differentiable at x=0
3) not differentiable at x= 1/2
4) continuous but not differentiable at x=0

Differentiability of functional equations:

13. Let f(x+y)=f(x)f(y) and f(x)=1+xg(x)G(x),

where $\lim_{x\to 0} g(x)=a$ **and** $\lim_{x\to 0} G(x)=b$. **Then** $f'(x)$ **is equal to**

1) 1+ab 2) ab 3) f(x) 4) abf(x)

14. **If** $f:R\to R$ **be a differentiable function, such that** $f(x+2y)=f(x)+f(2y)+4xy$ **for all** $x,y\in R$ **then**

1) $f'(1)=f'(0)+1$ 2) $f'(1)=f'(0)-1$

3) $f'(0)=f'(1)+2$ 4) $f'(0)=f'(1)-2$

15. **Let f be a differentiable function satisfying the condition** $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$, **for all** $x,y(\neq 0)\in R$ **and** $f(y)\neq 0$. **If** $f'(1)=2$, **then** $f'(x)$ **is equal to**

1) $2f(x)$ 2) $\frac{f(x)}{x}$ 3) $2xf(x)$ 4) $\frac{2f(x)}{x}$

LEVEL-II (C.W)-KEY

1) 3	2) 1	3) 1	4) 1	5) 2
6) 3	7) 3	8) 1	9) 3	10) 1
11) 3	12) 3	13) 4	14) 4	15) 4

LEVEL-II (C.W)-HINTS

1. $f'(x)=\begin{cases}-x & \text{if } x<1\\ 3x & \text{if } x\geq 1\end{cases}$

2. $f'(0^+)=$

$$\lim_{x\to 0^+}\left(\frac{f(x)-f(0)}{x-0}\right)=\lim_{x\to 0^+}\left(\frac{\frac{x\sqrt{x}}{x+2}-0}{x-0}\right)$$

3. fog=I $\Rightarrow fog(x)=x$ for all x $\Rightarrow f'(g(x))g'(x)=1$ for all x

$$\Rightarrow f'(g(a))=\frac{1}{g'(a)}=\frac{1}{2}\Rightarrow f'(b)==\frac{1}{2}\ (\because g(a)=b)$$

4. f(x)= $\begin{cases}-x^3, & \text{if } x<0\\ 0, & \text{if } x=0\\ x^3, & \text{if } x>0\end{cases}$ and $f'''(x)=\begin{cases}-6, & \text{if } x<0\\ 0, & \text{if } x=0\\ 6, & \text{if } x>0\end{cases}$

$f'''(0-)\neq f'''(0+)$ $f'''(x)$ is exists except at x=0

$\therefore$ number of points = ∞

5. For $-\pi/2<x\leq 0$, f(x)= $-p\sin x+qe^{-x}-rx^3$, so

= -p -q

For $0<x<\pi/2$, f(x)=p sin x+qe^x+rx^3,

$$f'(0+)=\lim_{x\to 0+}\frac{f(x)-f(0)}{x-0}=\lim_{x\to 0+}\left[\frac{p\sin x}{x}+q\left(\frac{e^x-1}{x}\right)-rx^2\right]=p+q$$

For f to be diffentiable at x=0,we must have

p+q = - p -q $\Rightarrow$ p + q =0.

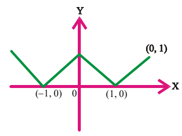
6. $f'(x)=\begin{cases}-3, & \text{if } x<1\\ x, & \text{if } x\geq 1\end{cases}$

7. We have $\frac{\sin 4\pi[x]}{1+[x]^2}=0,\ \forall x$

[$\because 4\pi[x]$ is an integral multiple of 'π']

$\Rightarrow f(x)=0$ for all x

8.



9. $f:R\to R$ **is an even function**

$f'(x)$ **is an odd function**

Hence $f''(x)$ **is even**

$f''(-\pi)=f''(\pi)=1$

10. When $t\geq 0$, we have

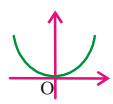
$x=2t-t$ and

$y=t^2+t^2=2t^2\Rightarrow y=2x^2, x\geq 0$

When t<0, we have

x = 2t +t = 3t and y = t^2 - t^2 = 0 $\Rightarrow$ y =0 for all x <0

11.



$$|f(x)| = \begin{cases} 1, & -2 \le x \le 0 \\ 1-x^2, 0 < x \le 1 \\ x^2-1, 1 < x \le 2 \end{cases}$$ and $f|(x)| = x^2-1, -2 \le x \le 2$

$$\therefore g(x) = \begin{cases} x^2, & -2 \le x \le 0 \\ 0, & 0 < x \le 1 \\ 2(x^2-1), & 1 < x \le 2 \end{cases}$$

(by adding the function in proper domains)

$\therefore g(x)$ is differentiable everywhere except at x=1

12. We have, $f(x) = \begin{cases} [\cos \pi x] & x < 1 \\ |x-2| & 1 < x < 2 \end{cases}$

$$= \begin{cases} 2-x, & 1 \le x < 2 \\ -1, & 1/2 < x < 1 \\ 0, & 0 < x \le 1/2 \\ 1, & x = 0 \\ 0, & -1/2 \le x < 0 \\ -1, & -1/2 < x < -1/2 \end{cases}$$

It is evident from the definition that f(x) is discontinuous at x=1/2

13. $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

$[\because f(x+y) = f(x)f(y)]$

$= f(x) \lim_{h \to 0} \frac{f(h)-1}{h} = f(x) \lim_{h \to 0} \frac{1+hg(h)G(h)-1}{h}$

$= f(x) \lim_{h \to 0} g(h)G(h) = f(x) \lim_{h \to 0} G(h) \lim_{h \to 0} g(h) = abf(x)$

14. $f(x+2y) = f(x) + f(2y) + 4xy$ for all $x, y \in R$ putting $x = y = 0$, we get $f(0) = 0$

Now, $f(x+2y) = f(x) + f(2y) + 4xy$

$$\Rightarrow \frac{f(x+2y)-f(x)}{2y} = 2x + \frac{f(2y)}{2y}$$

$$\Rightarrow \lim_{y \to 0} \frac{f(x+2y)-f(x)}{2y}$$

$$= \lim_{y \to 0} \left\{ 2x + \frac{f(2y)-f(0)}{2y} \right\}$$

$\Rightarrow f'(x) = 2x + f'(0)$ for all x

$\Rightarrow f'(1) = 2 + f'(0)$

15. $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$, replacing x and y both by 1, we get

$$f(1) = \frac{f(1)}{f(1)} \Rightarrow f(1) = 1$$

Now $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

$$\Rightarrow f'(x) = f(x) \lim_{h \to 0} \left\{ \frac{\frac{f(x+h)}{f(x)} - 1}{h} \right\}$$

$$\Rightarrow f'(x) = f(x) \lim_{h \to 0} \left\{ \frac{f\left(\frac{x+h}{x}\right) - 1}{h} \right\}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} \lim_{h \to 0} \frac{f\left(1+\frac{h}{x}\right) - f(1)}{\frac{h}{x}}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} f'(1) = \frac{2f(x)}{x}$$

LEVEL -II (H.W)

Differentiability at a point:

1. If $f(x) = |x| + |\sin x|$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then its left hand derivative at $x = 0$ is **(Eam-2011)**

1) 0 2) -1 3) -2 4) -3

2. **The function**

$$f(x) = (x^2-1)|x^2-3x+2| + \cos(|x|)$$

is not differentiable at

1) x=-1 2) x=0 3) x=1 4) x=2

3. Let $f(x)=\begin{cases}\dfrac{\sin|x^2-5x+6|}{x^2-5x+6}, & x\neq 2,3\\ 1, & x=2 \text{ or } 3\end{cases}$ the set of all points where f is differentiable is

1) $(-\infty,\infty)$ 2) $(-\infty,\infty)\sim\{2\}$
3) $(-\infty,\infty)\sim\{3\}$ 4) $(-\infty,\infty)\sim\{2,3\}$

4. $f(x)=|\cos x|$ is not differentiable for the points given by $x=$

1) $\frac{\pi}{2}$ 2) $(2n+1)\pi,\ \forall n\in I$
3) $(2n+1)\frac{\pi}{2}\ \forall n\in I$ 4) 0

5. Let 'f' be a function defined by $2f(\sin x)+f(\cos x)=x\forall x$, then set of points where 'f' is not differentiable is

1) Set of all natural numbers
2) Set of all irrational numbers
3) $\{0,1,-1\}$ 4) $\{1,-1\}$

6. If $f(x)=\begin{cases}(\cos x)^{1/\sin x} & for\ x\neq 0\\ k & for\ x=0\end{cases}$ the value of k, so that f is differentiable at x=0 is

1) 0 2) 1 3) ½ 4) 2

7. If 0<a<1 and f is defined as

$f(x)=\begin{cases}x^a \text{ if } x>0\\ 0 \text{ if } x=0\end{cases}$ then

1) f is differentiable at x=0
2) f is not continuous
3) f is not differentiable at x=0
4) f is bounded

8. If f(x)= $\begin{cases}1, & x<0\\ 1+\sin x, & 0\le x<\pi/2\end{cases}$, then derivative of f(x) at x=0

1) is equal to 1 2) is equal to 0
3) is equal to -1 4) does not exist

Differentiability over an interval:

9. Let h(x)=min{x,x²} for $x\in\mathbb{R}$. Then which of the following is correct

1) h is continuous for all x
2) h is differentiable for all x
3) h'(x)=1 for all x>1
4) h is not a differentiable at 2 values of x

10. Let $f(x)=\begin{cases}3^x & \forall|x|\le 1\\ 7-x & \forall 1<x<7\end{cases}$ then f(x) is

1) continuous $\forall\ 1\le x\le 7$ but not differentiable at x=1
2) continuous $-1\le x<7$ & differentiable at x=1
3) neither continuous in [-1,7) nor differentiable at x=1
4) continuous & differentiable at x=1

Differentiability of functional equations:

11. If $f(x+y)=2f(x)f(y)$ all $x,y\in R$ where $f'(0)=3$ and $f(4)=2$, then $f'(4)$ is equal to

1) 6 2) 12 3) 4 4) 3

12. Let f(x+y)=f(x)f(y) and f(x)=1+(sin2x)g(x) where g(x) is continuous. Then $f^1(x)$ equals

1) f(x)g(0) 2) 2f(x)g(0)
3) 2g(0) 4) 2f(0)

LEVEL-II (H.W)-KEY

1) 3	2) 4	3) 4	4) 3	5) 4
6) 2	7) 3	8) 4	9) 4	10) 3
11) 2	12) 2			

LEVEL-II (H.W)-HINTS

1. $x\to 0^-, f(x)=-x-\sin x$

2. By verification $f'(2-)\neq f'(2+)$
$\therefore$ f(x) is not differentiable at x=2

3. The function is clearly differentiable except possible at x =2,3

$f'(2+)=\lim_{h\to 0+}\frac{f(2+h)-f(2)}{h}=\lim_{h\to 0+}\frac{\sin h(1-h)+h(1-h)}{h^2(-1+h)}$

$=-\lim_{h\to 0+}\left(\frac{\sin h(1-h)}{h(1-h)}+1\right)\frac{1}{h}$ which is does not exist

4. f is not differentiable at all points where cosx = 0

5. $2f(\sin x)+f(\cos x)=x$ (1)

Replacing 'x' by $\frac{\pi}{2}-x$,

$2f(\cos x)+f(\sin x)=\frac{\pi}{2}-x$ (2)

Solving (1) and (2) $3f(\sin x)=3x-\frac{\pi}{2}$

(or) $f(\sin x)=x-\frac{\pi}{6}$

$\therefore f(x) = Sin^{-1}x - \frac{\pi}{6}$

$f'(x) = \frac{1}{\sqrt{1-x^2}}$ is not differentiable at x = 1,-1

6. $\lim_{x\to 0}(\cos x)^{1/\sin x} = k$

$k = e^{\lim_{x\to 0}\left[\frac{1}{\sin x}\log(\cos x)\right]} = e^{\lim_{x\to 0}\frac{-\tan x}{\cos x}} = e^0 = 1$

7. $f'(0+) = \lim_{x\to 0}\frac{x^a - 0}{x-0} = \lim_{x\to 0}\frac{1}{x^{1-a}}$

$0<a<1 \Rightarrow (1-a)>0 \Rightarrow f'(0) = \infty$, does not exist

8. We have (LHD at x=0)=$\left[\frac{d}{dx}(1)\right]_{x=0} = 0$

9. $h(x) = \begin{cases} x, & x \ge 1 \\ x^2, & 0 \le x < 1 \\ x, & x < 0 \end{cases}$

From the graph it is clear that h is continuous. Also h is differentiable except possible at x=0 and 1

$h'(x) = \begin{cases} 1, & x > 1 \\ 2x, & 0 < x < 1 \\ 1 & x < 0 \end{cases}$

for x=1, $h'(1+) = \lim_{t\to 0+}\frac{h(1+t)-h(1)}{t} = \lim_{t\to 0+}\frac{1+t-1}{t} = 1$

but $h'(1-) = \lim_{t\to 0+}\frac{h(1-t)-h(1)}{t} = \lim_{t\to 0+}\frac{(1-t)^2-1}{t} = -2$

so h is not differentiable at 1

similarly $h'(0+) = 0$ but $h'(0-) = 1$

10. $f(x) = \begin{cases} 3^x & \forall |x| \le 1 \\ 7-x & 1 < x < 7 \end{cases}$

$f'(1^-) = 3\log 3$, $f'(1+) = 0-1=-1$ $f'(1-) \ne f'(1+)$

$\therefore$ f(x) is not differentiable at x=1

Hence f is not differentiable at x=1

11. $f'(4) = \lim_{h\to 0}\frac{f(4+h)-f(4)}{h}$

$\Rightarrow f'(4) = \lim_{h\to 0}\frac{f(4+h)-f(4+0)}{h}$

$\Rightarrow f'(4) = \lim_{h\to 0}\frac{2f(4)f(h)-2f(4)f(0)}{h}$

$\Rightarrow f'(4) = \lim_{h\to 0} 2f(4)\left[\frac{f(h)-f(0)}{h-0}\right]$

$f'(4) = 4\lim_{h\to 0}\frac{f(h)-f(0)}{h-0}$

$= 4f'(0) = 4\times 3 = 12$

12. $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f(x)f(h)-f(x)}{h}$

$= f(x)\lim_{h\to 0}\frac{f(h)-1}{h} = f(x)\lim_{h\to 0}\frac{1+(\sin 2h)g(h)-1}{h}$

$= f(x)\lim_{h\to 0}\frac{\sin 2h}{h}.\lim_{h\to 0}g(h) = 2f(x)g(0)$

LEVEL - III

1. **If** $f(x) = |\cos x - \sin x|$**, then** $f'\left(\frac{\pi}{4}\right) =$

1) $\sqrt{2}$ **2)** $-\sqrt{2}$

3) 0 **4) does not exists**

2. **Let f(x) be a polynomial of degree two which is positive for all x $\in$ R.**

$g(x) = f(x) + f'(x) + f''(x) + f'''(x) + x^2 f^{iv}(x)$,

then for any real x

1) g(x)<0 2) g(x)>0 3) g(x)=0 4) g(x) $\ge$ 0

3. **Let f:R $\to$ R & g:R $\to$ R be defined by g(x)=xf(x) then**

1) g is a differentiable function

2) g is differentiable at 0 if f is continuous at 0

3) g is a constant function

4) f is a differentiable function

4. **Let f(x)=|[x]x|, for -1 $\le$ x $\le$ 2 Where [] represents the integral part function then**

1) f(x) is discontinuous at x=0

2) f(x) is differentiable at x=1

3) f(x) is not differentiable at x=2

4) f(x) is differentiable at x=2

5. The function f(x)=|x³| is

1) differentiable everywhere
2) continuous but not differentiable at x=0
3) not a continuous function
4) a function with range $(0,\infty)$

6. Which of the following function is differentiable at *x=0*

1) $cos(|x|)+|x|$ 2) $cos(|x|)-|x|$
3) $sin(|x|)+|x|$ 4) $sin(|x|)-|x|$

7. The set of all points differentiability of the function f(x)=$\frac{\sqrt{x+1}-x}{\sqrt{x}}$ for $x\neq 0$ and f(0)=0 is

1) $(-\infty,\infty)$ 2) $[0,\infty)$ 3) $(0,\infty)$ 4) $(-\infty,\infty)\sim\{0\}$

8. For what triplets of real numbers (a,b,c) with $a\neq 0$ the function f(x)

$$=\begin{cases} x & x\le 1 \\ ax^2+bx+c & otherwise \end{cases}$$

is differentiable for all real x

1) $\{(a,1-2a,a)/a\in R, a\neq 0\}$
2) $\{(a,1-2a,c)/a,c\in R, a\neq 0\}$
3) $\{(a,b,c)/a,b,c\in R, a+b+c=1\}$
4) $\{(a,1-2a,0)/a\in R, a\neq 1\}$

9. Let f(x)=$\frac{|x|\left(3e^{1/|x|}+4\right)}{2-e^{1/|x|}}$, $x\neq 0$ and f(0)=0 then

1) f is not continuous
2) f is continuous but not differentiable at x=0
3) $f'(0)$ exist 4) $f'(0+)=2$

10. Given that f(x) is a differentiable function of x and that f(x).f(y)=f(x)+f(y)+f(xy)-2 and that f(2)=5. Then $f^1(3)$ is equal to

1) 6 2) 24 3) 15 4) 19

11. If f(x+y+z) = f(x).f(y).f(z) for all x,y,z and f(2)=5, $f(0)$=3, then $f'(2)$ equals

1) 15 2) 9 3) 16 4) 6

12. Suppose that f is a differentiable function with the property that f(x+y) =f(x)+f(y)+xy and $\lim_{h\to 0}\frac{1}{h}f(h)=3$, then

1) f is a linear function 2) $f(x)=3x+x^2$
3) $f(x)=3x+x^2/2$ 4) $f(x)=3x-x^2/2$

13. if $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}\ \forall x,y\in R$

and $f'(0)=-1, f(0)=1, then\ f(2)=$

1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) 1 4) -1

14. Let f(x) be differentiable function such that $f\left(\frac{x+y}{1-xy}\right)=f(x)+f(y)$ $\forall$ **x and y. If** $\lim_{x\to 0}\frac{f(x)}{x}=\frac{1}{3}$ **then** $f'(1)$ **equals**

1) $\frac{1}{4}$ 2) $\frac{1}{6}$ 3) $\frac{1}{12}$ 4) $\frac{1}{8}$

15. Let f(x) be defined for all x>0 and be continuous. Let f(x) satisfy $f\left(\frac{x}{y}\right)$=f(x)-f(y) for all x,y and f (e)=1 Then which of the following may be true

1) f(x) is bounded 2) $f(1/x)\to 0$ as $x\to\infty$
3) $x f(x)\to 1$ as $x\to 0+$ 4) $f'(x)=\frac{1}{x}$

16. Let f(x)=$\begin{cases} A+\sin^{-1}(x+B), & \forall x\ge 1 \\ x, & \forall x<1 \end{cases}$ is differentiable then

1) A=-1,B=-1 2) A=1,B=-1
3) A=B=1 4) A=0,B=1

17. Let f(x)=$\begin{cases} x^2-1, & x\le 1 \\ k(x-1), & x>1 \end{cases}$ then

1) f is continuous for only finitely many values of k
2) f is discontinuous at x=1
3) f is differentiable only when k=2
4) there are infinitely many values of k for which

f is differentiable

18. **Let f(x)=** $\begin{cases} x^n \sin\frac{1}{x}, & x \neq 0 \\ 0 & , x=0 \end{cases}$**, then f(x) is continuous but not differentiable at x=0 if**

1) $n \in (0,1]$ 2) $n \in [1,\infty)$
3) $n \in (-\infty, 0)$ 4) n=0

19. **Let f(x)=** $\begin{cases} \int_0^x \{1+|1-t|\}dt & \text{if } x>2 \\ 5x-7 & \text{if } x \le 2 \end{cases}$**, then**

1) f is not continuous at x=2
2) f is continuous but not differentiable at x=2
3) f is differentiable everywhere
4) $f'(2+)$ does not exist

20. **The values of a and b such that the function f defined as** $f(x)=\begin{cases} ax^2-b, & |x|<1 \\ -1/|x|, & |x| \ge 1 \end{cases}$ **is differentiable are**

1) a=1,b=-1 2) a=$\frac{1}{2}$,b=$\frac{1}{2}$
3) a=$\frac{1}{2}$,b=$\frac{3}{2}$ 4) a=$\frac{3}{2}$,b=$\frac{3}{2}$

21. **Let the function** $f(x)=\begin{cases} \frac{\sin^3 x^2}{x} & \forall x \neq 0 \\ 0 & \forall x=0 \end{cases}$

1) is continuous but not derivable at x=0
2) neither continuous nor differentiable at x=0
3) continuous but not differentiable at x=1
4) continuous and differentiable at x=0

22. **The function f(x)=** $\begin{cases} x^4+x^2-x+2 & \text{for } x \le 1 \\ 3x^3-x^2+x & \text{for } x>1 \end{cases}$ **is**

1) continuous everywhere
2) differentiable everywhere
3) differentiable at x=1
4) such that f' exists everywhere but f'' is not continuous at x=1

23. **Let f(x) be defined by** $f(x)=\begin{cases} \sin 2x & \text{if } 0<x\le \pi/6 \\ ax+b & \text{if } \pi/6<x\le 1 \end{cases}$

The values of a and b such that f and f' are continuous, are

1) a = 1, b = $1/\sqrt{2} + \pi/6$ 2) a = $1/\sqrt{2}$, b = $1/\sqrt{2}$
3) a = 1, b = $\sqrt{3}/2 - \pi/6$ 4) a = $\sqrt{3}/2$, b = $\sqrt{3}/2 + \pi/6$

24. $f(x)=\begin{cases} b\sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2}<x<0 \\ \frac{1}{2} & at\, x=0 \\ \frac{e^{\frac{ax}{2}}-1}{x} & 0<x<\frac{1}{2} \end{cases}$

If f(x) is differentiable at x =0 and $|c|<\frac{1}{2}$ then

1) a = 1 and $64b^2 + c^2 = 4$
2) a = 0 and $64b^2 + c^2 = 2$
3) a = 2 and $64b^2 + c^2 = 1$
4) a = 3 and $64b^2 + c^2 = 3$

25. **The function f defined by**

$f(x)=\begin{cases} \frac{\sin x^2}{x} & for\ x\neq 0 \\ 0 & for\ x=0 \end{cases}$ **is**

1) continuous and derivable at x=0
2) neither continuous nor derivable at x=0
3) continuous but not derivable at x=0
4) continuous but not derivable at x=1

26. **If f(x)=** $\begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)}, & x\neq -2 \\ 2 & , x=-2 \end{cases}$**, then f(x) is**

1) continuous at x = -2
2) not coninuous at x = -2
3) differentiable at x = -2
4) continuous but not derivable at x = -2

27. **If a function f(x) is defined as f(x)=** $\begin{cases} \frac{x}{\sqrt{x^2}}, & x\neq 0 \\ 0 & , x=0 \end{cases}$ **then**

1) f(x) is continuous at x=0 but not differentiable at x=0
2) f(x) is continuous as well as differentiable at x=0
3) f(x) is not continuous at x=0
4) f(x) is continuous everywhere

28. **At the point x=1, the function f(x) =** $\begin{cases} x^3-1; & \text{for } 1<x<\infty \\ x-1; & \text{for } -\infty<x\le 1 \end{cases}$ **is**

1) continuous and differentiable
2) continuous and not differentiable
3) discontinuous and differentiable

4) discontinuous and not differentiable

29. Let $f(x)=\begin{cases}(x-1)\sin\frac{1}{x-1}, & if\ x\neq 1\\ 0 & ,if\ x=1\end{cases}$

Then which one of the following is ture? [AIEEE 2008]

1) f is neither differentiable at $x=0$ nor at $x=1$
2) f is differentiable at $x=0$ and at $x=1$
3) f is differentiable at $x=0$ but not at $x=1$
4) f is differentiable at $x=1$ not at $x=0$

30. **If $\lim\limits_{x\to c}\frac{f(x)-f(c)}{x-c}$ exists finitely, then**

1) $\lim\limits_{x\to c}f(x)=f(c)$ 2) $\lim\limits_{x\to c}f'(x)=f'(c)$
3) $\lim\limits_{x\to c}f(x)$ does not exist
4) $\lim\limits_{x\to c}f(x)$ may or may not exist

31. **Let f(x) be a function differentiable at x=c. Then $\lim\limits_{x\to c}f(x)$ equals**

1) $f'(c)$ 2) $f''(c)$ 3) $\frac{1}{f(c)}$ 4) f(c)

32. **If x+4|y|=6y, then y as a function of x is**

1) continuous at x=0 2) derivable at x=0
3) $\frac{dy}{dx}=\frac{1}{2}$ for all x 4) $\frac{dy}{dx}$=0 for all x

33. **The set onto which the derivative of the function f(x)=x(logx -1) maps the set $[1,\infty)$ is**

1) $[1,\infty)$ 2) $(0,\infty)$ 3) $[0,\infty)$ 4) (0,0)

34. **Let g(x)=$\begin{cases}e^{2x} & \forall x<0\\ e^{-2x} & \forall x\geq 0\end{cases}$ then g(x) does not satisfies the condition**

1) continuous $\forall x\in R$
2) not differentiable at x=0
3) continuous $\forall x\in R$ & non differentiable at x=0
4) g(x) is continuous & differentiable everywhere

35. **If the function f(x)=$\left[\frac{(x-5)^3}{A}\right]$sin(x-5) +acos(x-2), where [.] denotes the greatest integer function and a$\in$R ,is continuous and differentiable in (7,9) then**

1) A $\in$ [8,64] 2) A $\in$ (0,8]
3) A $\in$ $[64,\infty)$ 4) A $\in$ (0,0)

36. **Let f(x)=$[x]^2+\sqrt{\{x\}}$,where [ ] & { } respectively denotes the greatest integer and fractional part of functions, then**

1) f(x) is continuous at all integral points
2) f(x) is not differentiable $\forall x\in I$
3) f(x) is discontinuous as $x\in I-\{1\}$
4) f(x) is continuous & differentiable at x=0

37. **If f(x)=$\sqrt{x+2\sqrt{2x-4}}+\sqrt{x-2\sqrt{2x-4}}$ then f(x) is differentiable on**

1) $(-\infty,\infty)$ 2) $[2,\infty)-\{4\}$ 3) $[2,\infty)$ 4) $(0,\infty)$

38. **Let f(x)=(x+|x|)|x|, then for all x**

1) f is continuous for some x
2) f is differentiable for some x
3) f' is continuous
4) f'' is continuous

39. **If f(x)=$x\left(\sqrt{x}-\sqrt{x+1}\right)$ then**

1) f is continuous but not differentiable at x=0
2) f is differentiable at x=0
3) f is differentiable but not continuous at x=0
4) f is not differentiable at x=0

40. **The number of points at which the function f(x)=|x-0.5|+|x-1|+tanx does not have a derivative in the interval (0,2) is**

1) 1 2) 2 3) 3 4) 4

41. **Suppose $f(x)$ is differentiable at $x=1$ and $\lim\limits_{h\to 0}\frac{1}{h}f(1+h)=5$, then $f'(1)$ equals**

1) 3 2) 4 3) 5 4) 6

42. **The set of poits where $f(x)=\frac{x}{1+|x|}$ is differentiable is**

1) $(-\infty,0)\cup(0,\infty)$ 2) $(-\infty,-1)\cup(-1,\infty)$

3) $(-\infty,\infty)$ 4) $(0,\infty)$

43. **Let $f:R\to R$ be a funtction defined by $f(x)=\min\{x+1,|x|+1\}$, Then which of the following is true? [AIEEE - 2007]**

1) $f(x)$ is differentiable everywhere
2) $f(x)$ is not diffferentiable at $x=0$
3) $f(x)\geq 1$ for all $x\in R$
4) $f(x)$ is not differentaible at $x=1$

44. **If funcion $f(x)$ is differentiable at $x=a$, then $\lim\limits_{x\to a}\dfrac{x^2f(a)-a^2f(x)}{x-a}$ is : [AIEEE- 2011]**

1) $-a^2f'(a)$
2) $a\,f(a)-a^2f'(a)$
3) $2a\,f(a)-a^2f'(a)$
4) $2a\,f(a)+a^2f'(a)$

45. **If $f:(-1,1)\to R$ be a differentiable function with $f(0)=-1$ and $f'(0)=1$. Let $g(x)=[f(2f(x)+2)]^2$, then $g'(0)$ is equal to (AIEEE-2010)**

1) 4 2) -4 3) 0 4) -2

LEVEL-III - KEY

1) 4	2) 2	3) 2	4) 3	5) 1
6) 4	7) 4	8) 1	9) 2	10) 1
11) 1	12) 3	13) 4	14) 2	15) 4
16) 2	17) 3	18) 1	19) 2	20) 3
21) 4	22) 1	23) 3	24) 1	25) 1
26) 2	27) 3	28) 2	29) 3	30) 1
31) 4	32) 1	33) 3	34) 3	35) 3
36) 3	37) 2	38) 3	39) 2	40) 3
41) 3	42) 3	43) 1	44) 3	45) 2

LEVEL-III - HINTS

1. $f(x)=\begin{cases}\cos x-\sin x & \text{for } x\in(0,\pi/4)\\ \sin x-\cos x & \text{for } x\in(\pi/4,\pi/2)\end{cases}$

2. Let $f(x)=ax^2+bx+c$. As $f(x)>0$ for all $x\in R$, we must have, $a>0$ and $b^2-4ac<0$

$g(x)=ax^2+bx+c+(2ax+b)+2a+0+(x^2).0$

$=ax^2+(b+2a)x+b+c+2a$

Discriminant of $g(x)=(b+2a)^2-4a(b+c+2a)$

$=-4a^2+(b^2-4ac)<0$

Thus $g(x)>0$ for $x\in R$

3. By options (2) is correct.

4. $f(x)=\begin{cases}-x, & -1\le x<0\\ 0, & 0\le x<1\\ x, & 1\le x<2\\ 4, & 2\le x<3 \text{ or } x=2(\text{as given})\end{cases}$

Now $\underset{x\to0-}{Lt}f(x)=0$, $\underset{x\to0+}{Lt}f(x)=0$ then $f(0)=0$

$\therefore$ f(x) is continuous at x=0

Again $\underset{x\to1-}{lt}f(x)=0$, $\underset{x\to1+}{lt}f(x)=\underset{x\to1}{lt}x=1$

f(x) is discontinuous at x=1 so we say f(x) is non-differcutiavle at x =1

Again $\underset{x\to2-}{lt}f(x)=\underset{x\to2}{lt}x=2\neq(f(x))_{x=2}$ i.e $\neq 4$

$\therefore$ f(x) is discontinuous at x=2)

So non-differentiable at x=2

5. $f(x)=|x^3|$ then $f(x)=\begin{cases}-x^3 & \text{if } x<0\\ 0 & \text{if } x=0\\ x^3 & \text{if } x>0\end{cases}$

$f'(0-)=0; f'(0+)=0\Rightarrow f'(0-)=f'(0+)$

then f is differentiable at x=0

i.e f is continuous at x=0

6. *$\cos|x|=\cos x$ is differentialbe at x=0, but $|x|$ is not differentiable at x=0, Hence 1 & 2 options are not correct.*

$f(x)=\sin|x|-|x|=\begin{cases}-\sin x+x & x<0\\ \sin x-x & x\geq0\end{cases}$

is differentiable at x=0

7. $f'(x)$ is not exist at x=0

$\therefore$ f(x) is differentiable $\forall x\in(-\infty,\infty)-\{0\}$

8. $f'(x)=\begin{cases}1 & x\leq1\\ 2ax+b & x>1\end{cases}$

f is differentiable, then $f'(1^+) = f'(1^-)$

$\Rightarrow 1 = 2a + b \Rightarrow b = 1 - 2a ----(1)$

Given f(x) is continous at x =1 then 1 =a+b+c

9. $f'(0-) = \lim_{x\to 0-} \dfrac{\dfrac{|x|\left(3e^{1/|x|}+4\right)}{2-e^{1/|x|}} - 0}{x-0} = \lim_{x\to 0} \dfrac{\dfrac{-x\left(3e^{-1/x}+4\right)}{2-e^{-1/x}} - 0}{x-0}$

$= \dfrac{3e^{-\infty}+4}{2-e^{-\infty}} = 2$

$f'(0+) = \lim_{x\to 0+} \dfrac{\dfrac{|x|\left(3e^{1/|x|}+4\right)}{2-e^{1/|x|}} - 0}{x-0} = \lim_{x\to 0} \dfrac{\dfrac{x\left(3e^{1/x}+4\right)}{2-e^{1/x}} - 0}{x-0}$

$= \lim_{x\to 0} \dfrac{\left(3+\dfrac{4}{e^{1/x}}\right)}{\dfrac{2}{e^{1/x}}-1} = \dfrac{3}{-1} = -3 \Rightarrow \quad f'(0-) \neq f'(0+)$

then f is not differentiable at x=0 and continuous at x=0

10. We have, $f(x).f(y) = f(x) + f(y) + f(xy) - 2$

$\Rightarrow f(x).f\left(\tfrac{1}{x}\right) = f(x) + f\left(\tfrac{1}{x}\right) + f(1) - 2$

$\Rightarrow f(x).f\left(\tfrac{1}{x}\right) = f(x) + f\left(\tfrac{1}{x}\right)$

(since $f(1)=2$ putting x=y=1)

$\Rightarrow f(x) = x^n + 1 \Rightarrow f(2) = x^2 + 1$ (since f(2)=5)

$\Rightarrow n = 2$

$\therefore f(x) = x^2 + 1 \Rightarrow f(3) = 10$

11. We have, $f(x+y+z) = f(x)f(y)f(z)$ for all x,y,z

$\Rightarrow f(0) = f(0)f(0)f(0)$ (putting x=y=z=0)

$\Rightarrow f(0)\left\{1-(f(0))^2\right\} = 0 \Rightarrow f(0) = 1$

($\because f(0)=0 \Rightarrow f(x)=0$ for all x)

Putting z=0 and y=2, we get

$f(x+2) = f(x)f(2)f(0) \Rightarrow f(x+2) = 5f(x)$ for all x

$\Rightarrow f'(2) = 5f'(0) = 5\times 3 = 15$

hence f is differentiable everywhere

12. $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \dfrac{f(x)+f(h)+xh-f(x)}{h}$

$= = \lim_{h\to 0} \dfrac{1}{h} f(h) + x = 3 + x$

Hence f(x)=3x+ $x^2/2$ +c. Putting x=y=0 in the given equation, we have f(0)=f(0)+f(0)+0 $\Rightarrow$ f(0)=0. Thus c=0 and f(x)=3x+ $x^2/2$

13. Take *f(x) = ax+b*

14. $f\left(\dfrac{x+y}{1-xy}\right) = f(x) + f(y) \Rightarrow f(x) = A\tan^{-1} x$

Now, $\lim_{x\to 0} \dfrac{f(x)}{x} = A \lim_{x\to 0} \dfrac{\tan^{-1} x}{x}$, $\therefore A = 1/3 \quad \therefore f(x) = \dfrac{1}{3}\tan^{-1} x$

15. A function satisfying the given functional equation is f(x)=log x. Clearly f(x) is not bounded and $f(1/x) = -\log x \not\to 0$ as $x \to \infty$.

Also x f(x)=xlog x $\not\to$ 1 as $x \to 0$. But $f'(x) = 1/x$.

In fact, it can be shown that any continuous functions satisfying the given functional equation is of the form k log x, for some k>0.

Since f(e)=1 so k=1

16. As $x \geq 1$ $\sin^{-1}(x+B)$ is defined when B = -1 from the options.f is differentiable $\Rightarrow$ f is continuous.

17. $f(x) = \begin{cases} x^2-1 & \text{if } x \leq 1 \\ k(x-1) & \text{if } x > 1 \end{cases}$ then $f'(x) = \begin{cases} 2x & \text{if } x \leq 1 \\ k & \text{if } x > 1 \end{cases}$

$f'(1-) = f'(1+)$ i.e 2(1)=k then k=2

18. since f(x) is continuous at x=0, therefore

$\lim_{x\to 0} f(x) = f(0) = 0 \Rightarrow \lim_{x\to 0} x^n \sin\left(\tfrac{1}{x}\right) = 0 \Rightarrow n > 0$

f(x) is differentiable at x=0 if

$\lim_{x\to 0} \dfrac{f(x)-f(0)}{x-0}$ exists finitely

$\Rightarrow \lim_{x\to 0} \dfrac{x^n \sin(1/x) - 0}{x}$ exists finitely

$\Rightarrow \lim_{x\to 0} x^{n-1}\sin(1/x)$ exists finitely $\Rightarrow n-1 > 0 \Rightarrow n > 1$

If $n \leq 1$, then $\lim_{x\to 0} x^{n-1}\sin(1/x)$ does not exist and hence f(x) is not differentiable at x=0

hence f(x) is continuous but not differentiable at x=0 for $0 < n \leq 1$, i.e $n \in (0,1]$

19. For x>2, $\int_0^x \{1+|1-t|\}dt = \int_0^1 \{1+|1-t|\}dt + \int_1^x \{1+|1-t|\}dt$

$= \int_0^1 (2-t)dt + \int_1^x t\,dt = 1 + \dfrac{x^2}{2}$

Thus, $f(x) = \begin{cases} 1 + x^2/2, & x > 2 \\ 5x-7, & x \leq 2 \end{cases}$

$\lim_{x\to 2+} f(x) = 1 + 4/2 = 3 = f(2) = \lim_{x\to 2-} f(x)$

$f'(2+) = \lim_{h\to 0+} \dfrac{1+(1/2)(2+h)^2 - 3}{h} = \lim_{h\to 0+} \dfrac{(1/2)h^2 + 2h}{h} = 2$

$f'(2-) = \lim_{h\to 0+} \dfrac{5(2+h)-7-3}{h} = 5$

Hence f is continuous but not differentiable at x=2

20. Since every differentiable function is continuous, so we must have

$\lim_{x\to 1-} f(x) = f(1) \Rightarrow a-b=-1$

for f to be differentiable, $f'(1-) = f'(1+)$

$\Rightarrow \lim_{h\to 0-}\left[\frac{a(1+h)^2-b+1}{h}\right] = \lim_{h\to 0-}\left[\frac{-1|1+h|+1}{h}\right]$

$= \lim_{h\to 0-}\left[\frac{a(2h+h^2)}{h}\right] = \lim_{h\to 0+}\frac{h}{h(1+h)}$ (as a-b=-1)

$\Rightarrow 2a=1$, Hence $a=\frac{1}{2}$ and $b=\frac{3}{2}$

21. $\lim_{x\to 0} f(x) = \underset{x\to 0}{Lt}\frac{\sin x^2}{x^2}(x\sin^2 x^2) = \underset{x\to 0}{Lt}\left(\frac{\sin x^2}{x^2}\right).\underset{x\to 0}{Lt}(x\sin^2 x^2)$

= 0=f(0)

and LHD at x=0 is $\underset{h\to 0}{Lt}\frac{f(0+h)-0}{h} = \frac{\frac{\sin^3 h^2}{h}-0}{h}$ =0

RHD=0

$\therefore$ f(x) is differentiable at x=0

22. f(x)=$\begin{cases} x^4+x^2-x+2 & \text{for } x\le 1 \\ 3x^3-x^2+x & \text{for } x>1 \end{cases}$

$f'(x) = \begin{cases} 4x^3+2x-1 & \text{for } x\le 1 \\ 9x^2-2x+1 & \text{for } x>1 \end{cases}$

$f'(1-)=5, f'(1+)=8 \Rightarrow f'(1-)\ne f'(1+)$

$\therefore$ f(x) is not differentiable at x=1

$\lim_{x\to 1} f(x) = f(1)$ satisfied, then f is continuous at x=1

$\therefore$ f(x) is continuous at everywhere

23. $f(x) = \begin{cases} \sin 2x & \text{if } 0<x\le \pi/6 \\ ax+b & \text{if } \pi/6<x\le 1 \end{cases}$,

then $f'(x) = \begin{cases} 2\cos 2x & \text{if } 0<x\le \pi/6 \\ a & \text{if } \pi/6<x\le 1 \end{cases}$

f and f' are continuous

$\lim_{x\to \pi/6-} f(x) = \lim_{x\to \pi/6+} f(x) \Rightarrow \lim_{x\to \pi/6}\sin 2x = \lim_{x\to \pi/6} ax+b$

$\frac{\sqrt{3}}{2} = \frac{a\pi}{6}+b \Rightarrow b = \frac{\sqrt{3}}{2}-\frac{\pi}{6}$

$\lim_{x\to \pi/6-} f'(x) = \lim_{x\to \pi/6+} f'(x)$ then $\lim_{x\to \pi/6} 2\cos 2x = \lim_{x\to \pi/6} a$

i.e. a=1

24. Find $f'(x)$ and $f'(0^+) = f'(0^-)$

25. $f'(0) = \lim_{x\to 0}\frac{\frac{\sin x^2}{x}-0}{x-0} = 1$

then f is differentiable at x=0

26. $\lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h) = \lim_{h\to 0}\frac{|-2-h+2|}{\tan^{-1}(-2-h+2)}$

$= \lim_{h\to 0}\frac{h}{\tan^{-1}(-h)} = \lim_{h\to 0}\frac{-h}{\tan^{-1}(h)} = -1$

and $\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(-2+h) = \lim_{h\to 0}\frac{|-2+h+2|}{\tan^{-1}(-2+h+2)}$

$= \lim_{h\to 0}\frac{h}{\tan^{-1}(h)} = 1$ $\quad \therefore \lim_{x\to 2-} f(x) \ne \lim_{x\to 2+} f(x)$

so, f is neither continuous nor differentiable at x=-2

27. we have, $f(x) = \begin{cases} \frac{x}{\sqrt{x^2}}, & x\ne 0 \\ 0, & x=0 \end{cases} = \begin{cases} \frac{x}{|x|}, & x\ne 0 \\ 0, & x=0 \end{cases} = \begin{cases} 1, & x>0 \\ -1, & x<0 \\ 0, & x=0 \end{cases}$

f(x) is not continuous at x=0

28. we have, $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-}(x-1) = 0$ and

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1}(x^3-1) = 0$. Also $f(1) = 1-1 = 0$

so, f(x) is continuous at x=1)Clearly $Lf'(1)=2$ and $Rf'(1)=3$. Therefore, $f(x)$ is not differentiable at x=1

29. $Lf'(1) = \lim_{h\to 0}\frac{(1-h-1)\sin\left(\frac{1}{1-h-1}\right)-0}{-h}$

$= -\lim_{h\to 0}\sin\frac{1}{h}$ similiraly $Rf'(1) = \lim_{h\to 0}\sin\frac{1}{h}$

'f' is not differentiable at x=1, Clearly 'f' is differentiable at x =0.

As $Lf'(0) = Rf'(0) = Cos1 - Sin1$

30. since f(x) is differentiable at x=c, therefore it is continuous at x=c. Hence $\lim_{x\to c} f(x) = f(c)$

31. since f(x) is differentiable at x=c, therefore it is continuous at x=c. Hence $\lim_{x\to c} f(x) = f(c)$

32. We have, x+4|y|=6y

$\Rightarrow \begin{cases} x-4y=6y, & \text{if } y<0 \\ x+4y=6y, & \text{if } y\ge 0 \end{cases}$

$\Rightarrow y = \begin{cases} \frac{1}{2}x, & \text{if } x\ge 0 \\ \frac{1}{10}x, & \text{if } x<0 \end{cases} \Rightarrow y=f(x) = \begin{cases} \frac{1}{2}x, & \text{if } x\ge 0 \\ \frac{1}{10}x, & \text{if } x<0 \end{cases}$

clearly, $y=f(x)$ is continuous at x=0 but it is not differentiable at x=0

33. $f'(x) = \log x - 1 + x\left(\frac{1}{x}\right) = \log x$

since log x is increasing function so f' maps $[1,\infty)$

onto $[0,\infty)$

34. Given $g(x)=\begin{cases} e^{2x} & \forall\, x<0 \\ e^{-2x} & \forall\, x\ge 0 \end{cases}$, then

$$g'(x) = \begin{cases} 2e^{2x} & \forall\, x<0 \\ -2e^{-2x} & \forall\, x\ge 0 \end{cases}$$

$\therefore$ LHD at x=0 is $2e^0=2$

RHD at x=0 is $-2e^0=-2$

As LHD at x=0 $\ne$ RHD at x=0

$\therefore$ g(x) is not differentiable at x=0

Again $\underset{x\to 0-}{Lt}\, g(x) = \underset{x\to 0}{Lt}\, e^{2x} = 1$ and $\underset{x\to 0+}{Lt}\, g(x) = \underset{x\to 0}{Lt}\, e^{-2x} = 1$

$\therefore \underset{x\to 0-}{Lt}\, g(x) = \underset{x\to 0+}{Lt}\, g(x)$

$\therefore$ g(x) is continuous at x=0, more over g(x) is continuous $\forall\, x \in R$

35. [x] is not conitnuous and differentiable at integral values (points)

So f(x) is continuous and differentiable in (7,9)

if $\left[\dfrac{(x-5)^3}{A}\right]=0 \Rightarrow A\ge (9-5)^3 \Rightarrow A\ge 64 \therefore A\in[64,\infty)$

36. If $k\in I$

$\underset{x\to k+}{Lt}\, f(x)=k^2-0 \Rightarrow \underset{x\to k-}{Lt}\, f(x)=(k-1)^2+1$

Again $\underset{x\to k-}{Lt}\, f(x)=\underset{x\to k+}{Lt}\, f(x)=f(k)$

$k^2=(k-1)^2+1 \Rightarrow 2k=2$ then k=1

i.e., f(x) is continuous at k=1 and no other integral point.

So f(x) is discontinuous for all integral points except x=1

37. $f(x)=\sqrt{\left(\dfrac{\sqrt{2x-y}}{2}\right)^2+2+\sqrt{2x-4}}$

$+\sqrt{\dfrac{(2x-4)^2}{2}+2-2\sqrt{2x-4}}$

$\Rightarrow f(x)=\dfrac{1}{\sqrt{2}}\left[\left(\sqrt{2x\text{-}4}+2\right)+\left|\sqrt{2x-4}-2\right|\right]$

$\Rightarrow \left[1+\dfrac{dy}{dx}\right]=0 \qquad \therefore \dfrac{dy}{dx}=-1$

$$\Rightarrow f(x)=\begin{cases} \dfrac{4}{\sqrt{2}} \text{ if } \sqrt{2x-4}-2<0 \text{ i.e } \sqrt{2x-4}<2 \text{ i.e } x<4 \\ \sqrt{x-2} \text{ if } \sqrt{2x-4}\ge 2 \text{ i.e } 2x\ge 8 \text{ i.e } x\ge 4 \\ 2\sqrt{2} \text{ if } 2\le x<4 \end{cases}$$

Now $f'(x)=\dfrac{1}{2\sqrt{x-2}}$ if $4\le x<\infty$

0 if $2\le x\le 4$

But when x=4, $f'(4)=\dfrac{1}{0}$ not defined

$\therefore$ f(x) is differentiable on $[2,4)\cup(4,\infty)$

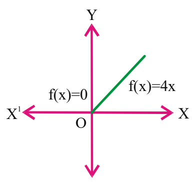
i.e $x\in[2,\infty)-\{4\}$

38. We have, $f(x)=\begin{cases} (x-x)(-x)=0, & x<0 \\ (x+x)x=2x^2, & x\ge 0 \end{cases}$

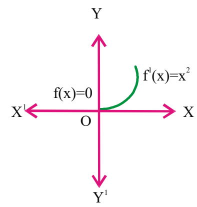
As is evident from the graph of f(x) that it is continuous and differentiable for all x, also

$f'(x)=\begin{cases} 0, & x<0 \\ 4x, & x\ge 0 \end{cases}$

clearly, $f'(x)$ continuous for all x but it is not differentiable at x=0



fig(i)



fig(ii)

39. we have $f(x)=x\left(\sqrt{x}-\sqrt{x+1}\right)$

$\Rightarrow f'(x)=\left(\sqrt{x}-\sqrt{x+1}\right)+\dfrac{x\left(\sqrt{x+1}-\sqrt{x}\right)}{2\sqrt{x}\sqrt{x+1}}$ for $x\ne 0,-1$.

Also $f'(0+)=-1$, so f(x) is differentiable and hence continuous at x=0

40. we have, $f(x)=|x-0.5|+|x-1|+\tan x, 0<x<2$

$$=\begin{cases} -2x+1.5+\tan x, & 0<x<0.5 \\ 0.5+\tan x, & 0.5\le x<1 \\ 2x-1.5+\tan x, & 1\le x<2 \end{cases}$$

It is evident from the above definition that $Lf'(0.5)\ne Rf'(0.5)$ and $Lf'(1)\ne Rf'(1)$. Also, the function is not continuous at $x=\pi/2$. So, it cannot be differentiable at three points

41. $f'(1)=\lim\limits_{h\to 0}\dfrac{f(1+h)-f(1)}{h}$

given that $\lim\limits_{h\to 0}\dfrac{f(1+h)}{h}=5$ and hence

$f(1)=0$

$f^1(1)=\lim\limits_{h\to 0}\dfrac{f(1+h)}{h}=5$

42. $f'(x)=\begin{cases}\frac{x}{(1-x)^2}; x<0\\ \frac{x}{(1+x)^2}; x\geq 0\end{cases}$

43. $f(x)=x+1, \forall x\in R$

44. $\lim_{x\to a}\frac{x^2f(a)-a^2f(x)}{x-a}$

$\lim_{x\to a}\frac{2xf(a)-a^2f'(x)}{1}$

$=2af(a)-a^2f'(a)$

45. $g'(x)=2\left[f(2f(x)+2)\right]$

$\times f'((2f(x))+2)\times 2f'(x)$

$g'(0)=\left[f(2f(0)+2)\right]$

$\times f'((2f(0))+2)\times 2f'(0)$

$=2f(0)\times f'(0)\times 2f'(0)$

$=2(-1)\times 1\times 2\times 1=-4$

LEVEL - IV

Here each question contains two statements :

Statement -1 (Assertion) and

Statement-2 (Reason):

each question has 4 choices (1), (2), (3) and (4) out of which, only one is correct

1) Statement-1 is true, statement -2 is true. Statement-2 is a correct exaplanation for statement-1

2) Statement-1 is true, statement -2 is true. Statement-2 is not a correct exaplanation for statement-1

3) Statement-1 is true, statement -2 is false.

4) Statement-1 is false, statement -2 is true.

1. **Statement-1:** if $y=f(x)=\frac{x}{1+|x|}, x\in R$, then f(x) is differentiable everywhere.

Statement-2: If $f(x)=\frac{x}{1+|x|}, x\in R$, then

$f'(x)=\begin{cases}\frac{1}{(1+x)^2}, x\geq 0\\ \frac{1}{(1-x)^2}, x\leq 0\end{cases}$

2. **Statement-1:** If

$f(x)=[\tan x], x\in\left[0,\frac{\pi}{3}\right]$, then

$f'\left(\frac{\pi}{4}\right)$ does not exist.

Statement-2: $Lf'\left(\frac{\pi}{4}\right)\to\infty$

3. **Statement-1:** $f(x)=x^n\sin\left(\frac{1}{x}\right)$ is differentiable for all real values of $x\,(n\geq 2)$.

Statement-2: For $n\geq 2$, right derivative = left derivative (for all real value of x)

4. **Statement-1:** The function $y=\sin^{-1}(\cos x)$ is not differentiable at the points where sin x=0.

Statement-2: $\frac{dy}{dx}=\frac{-\sin x}{|\sin x|}$, So the function is not differentiable at the points where sin x =0

5. **Statement-1:** $f(x)=e^{-|x|}$ is differentiable at x=0

Statement-2: $f(x)=e^{-|x|}$ is continuous at x=0

6. Let $f(x)=x|x|$ and g(x) = sin x

Statement-1: gof is differentiable at x=0 and its derivative is continuous at that point

Statement-2: gof is twice differentiable at x = 0

7. Consider the function,

$f(x)=|x-2|+|x-5|, x \in R.$

Statement-1: $f'(4)=0$

Statement-2: f is continuous in $[2,5]$, differentiable in $(2,5)$ and $f(2)=f(5)$.

LEVEL-IV - KEY

1) 1 2) 3 3) 1 4) 4 5) 4 6) 3 7) 1

LEVEL-IV - HINTS

1. verify at $x=0$

2. $$Rf' = \lim_{x\to\frac{\pi}{4}+} \frac{f(x)-f\left(\frac{\pi}{4}\right)}{x-\frac{\pi}{4}}$$

$$= \lim_{x\to\frac{\pi}{4}+} \frac{(\tan x)-0}{x-\frac{\pi}{4}} = \frac{1}{0} \to \infty$$

3. $x^n \sin\frac{1}{x}$ is differentiable when n>1

4. $$y=\sin^{-1}(\cos x)=\sin^{-1}\left(\sin\left(\frac{\pi}{2}+x\right)\right)$$

$$=\frac{\pi}{2}+x \Rightarrow \frac{dy}{dx}=1$$

$\therefore$ Statement-1 is false

statement-2 is clearly true

5. Statement-2:

$$\lim_{x\to 0-} f(x)=\lim_{x\to 0} e^{-(-x)}=\lim_{x\to 0} e^{x}=e^0=1=f(0)$$

statement-1:

$$Lf'(0)=\lim_{x\to 0-}\frac{f(x)-f(0)}{x-0}$$

$$=\lim_{x\to 0}\frac{e^{-(-x)}-1}{x}=\lim_{x\to 0}\frac{e^x-1}{x}=1$$

and $Rf'(0)$=-1

6. Let $h(x)=(gof)(x)$

$\therefore (gof)(x)=g(f(x))=\sin(x|x|)$

$$\therefore h(x)=\begin{cases}-\sin x^2 \text{ if } x<0\\ \sin x^2 \text{ if } x>0\end{cases}$$

$$\therefore h'(x)=\begin{cases}-2x\sin x^2 \text{ if } x<0\\ 2x\cos x^2 \text{ if } x>0\end{cases}$$

$$\therefore Rh'(0)=\lim_{x\to 0+}\frac{h(x)-h(0)}{x}$$

Now $$=\lim_{x\to 0}\frac{\sin x^2}{x}=0$$

7. $$f(x)=|x-2|=\begin{cases}x-2, x-2\ge 0\\ 2-x, x-2\le 0\end{cases}$$

$$=\begin{cases}x-2, x\ge 2\\ 2-x, x\le 2\end{cases}$$

Similarly, $$f(x)=|x-5|=\begin{cases}x-5, x\ge 5\\ 5-x, x\le 5\end{cases}$$

$$\therefore f(x)=|x-2|+|x-5|$$

$$=\{x-2+5-x=3, 2\le x\le 5\}$$

Thus $f\left(x\right)=3,2\le x\le 5$

$f'\left(x\right)=0,2<x<5$

$f'\left(4\right)=0,\therefore$ statement 1 is ture.

$\because f\left(2\right)=3$ and $\therefore f\left(5\right)=\left|5-2\right|+0=3$

$\therefore$ statement -2 is also true and a correct explantion for statement 1.

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